

Simultaneous equations and inequalities - Set 1 - Solutions

Specification MM1.4, MM1.5, M4.16, M4.17.

Answer key

Question	Answer	Topic
1	A	Simultaneous equations and inequalities
2	A	Simultaneous equations and inequalities
3	C	Simultaneous equations and inequalities
4	A	Simultaneous equations and inequalities
5	F	Simultaneous equations and inequalities
6	D	Simultaneous equations and inequalities
7	C	Simultaneous equations and inequalities
8	C	Simultaneous equations and inequalities
9	B	Simultaneous equations and inequalities
10	B	Simultaneous equations and inequalities
11	C	Simultaneous equations and inequalities
12	D	Simultaneous equations and inequalities

Worked solutions

Question 1 SIM-02 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.

Answer: A

$(2x + 1)(x - 3) \geq 0$: the roots are $-\frac{1}{2}$ and 3 and the parabola opens upwards, so $x \leq -\frac{1}{2}$ or $x \geq 3$.

Question 2 SIM-03 · Simultaneous equations and inequalities · spec MM1.5

Find the set of values of x satisfying both $3x - 7 < 2$ and $x^2 > 4$.

Answer: A

$3x < 9 \Rightarrow x < 3$. $x^2 > 4 \Rightarrow x < -2$ or $x > 2$. Combining: $x < -2$ or $2 < x < 3$.

Question 3 SIM-04 · Simultaneous equations and inequalities · spec MM1.4

The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

Answer: C

$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0$. Tangent when $4k^2 - 8(k^2 - 8) = 0 \Rightarrow -4k^2 + 64 = 0 \Rightarrow k = \pm 4$.

Question 4 SIM-05 · Simultaneous equations and inequalities · spec MM1.5

For how many integers n is $n^2 - 10n + 16 < 0$?

Answer: A

$(n - 2)(n - 8) < 0 \Rightarrow 2 < n < 8$, so $n = 3, 4, 5, 6, 7$: five integers.

Question 5 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

Answer: F

$x = y + 1$: $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. Solutions $(3, 2)$ with $x + y = 5$ and $(-2, -3)$ with $x + y = -5$. The sum is 0.

Question 6 SIM-07 · Simultaneous equations and inequalities · spec MM1.5

The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .

Answer: D

Need discriminant negative: $b^2 - 36 < 0$, so $-6 < b < 6$.

Question 7 SIM-09 · Simultaneous equations and inequalities · spec MM1.4

The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .

Answer: C

$x^2 - 2x + (c - 1) = 0$ has one solution when $4 - 4(c - 1) = 0$, so $c = 2$.

Question 8 SIM-12 · Simultaneous equations and inequalities · spec MM1.4

The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .

Answer: C

$$2x^2 - 2x - 4 - k = 0: \text{discriminant } 4 + 8(4 + k) > 0 \Rightarrow 36 + 8k > 0 \Rightarrow k > -\frac{9}{2}.$$

Question 9 SIM-14 · Simultaneous equations and inequalities · spec MM1.4

Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?

Answer: B

$$x^2 = y^2 \text{ means } y = x \text{ or } y = -x. \text{ } y = x: 5x = 12, \text{ one solution. } y = -x: -x = 12, \text{ one solution. Two in total.}$$

Question 10 SIM-17 · Simultaneous equations and inequalities · spec MM1.4

Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?

Answer: B

$$(x - y)^2 = 2(x^2 + y^2) - (x + y)^2 = 116 - 100 = 16, \text{ so the difference is 4 (the numbers are 3 and 7).}$$

Question 11 X2-14 · Simultaneous equations and inequalities · spec MM1.5

x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?

Answer: C

$$x^2 < 4 \text{ means } -2 < x < 2, \text{ which implies } x < 2 \text{ (sufficient); but } x = -5 \text{ satisfies } Q \text{ and not } P \text{ (not necessary).}$$

Question 12 SIM-01 · Simultaneous equations and inequalities · spec MM1.4

Solve the simultaneous equations $x + 2y = 7$ and $x^2 + y^2 = 10$. Find the sum of the possible values of x .

Answer: D

$$x = 7 - 2y: (7 - 2y)^2 + y^2 = 10 \Rightarrow 5y^2 - 28y + 39 = 0 \Rightarrow (5y - 13)(y - 3) = 0. \text{ } y = 3 \Rightarrow x = 1; y = \frac{13}{5} \Rightarrow x = \frac{9}{5}. \text{ Sum of } x \text{ values} = 1 + \frac{9}{5} = \frac{14}{5}.$$